

Xiaoyang Bai, Ph.D.

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Education

Ph.D., University of Illinois at Urbana-Champaign 2018 - 2023

- Major: Computer Science
- Advisor: Alexander Schwing

B.A., University of California, Berkeley 2014 - 2018

- Majors: Computer Science, Applied Math, Cognitive Science and Linguistics

Work Experience

Post-doctoral Fellow, The University of Hong Kong 2024-

- Affiliation: WeLight Lab at the Department of ECE
- Supervisors: Prof. Yifan (Evan) Peng

Graduate Research Assistant, University of Illinois at Urbana-Champaign 2018 – 2023

Research Scientist Intern, Amazon Studio LLC Summer 2022

AI Research Intern, Apple Inc. Summer 2020

Publications

PhyGaP: Physically-grounded Gaussians with polarization cues. Jiale Wu, **Xiaoyang Bai**^{*}, Zongqi He, Weiwei Xu, Yifan Peng^{*}. Accepted by *IEEE/CVF CVPR 2026* **Award Candidate**

Global underwater geolocation from time-lapsed polarization imagery. Sara Aghajanzadeh, **Xiaoyang Bai**, Zhongmin Zhu, David Forsyth, Viktor Gruev. Accepted by *IEEE/CVF CVPR 2026*

EventTracer: Fast path tracing-based event stream rendering. Zhenyang Li[†], **Xiaoyang Bai**[†], Jinfan Lu, Pengfei Shen, Edmund Y. Lam, Yifan Peng. In production for *IEEE Transaction on Visualization and Computer Graphics (TVCG)* 2026

ERF-GS: Reconstructing fast motion from disjoint event-RGB viewpoints. **Xiaoyang Bai**[†], Zhenyang Li[†], Weiwei Xu, Edmund Y. Lam, Yifan Peng. In production for *Computer Visual Media Journal (CVMJ)* (2026)

^{*} Corresponding authors.

[†] Equal contribution.

Structure-grounded training strategies aid generalization in stereo matching. Liangxun Ou, Yuhui Liu, Zhenyang Li, **Xiaoyang Bai**, Yifan Peng. Accepted by *the 13th International Conference on 3D Vision (3DV) 2026*

Enhanced velocity field modeling for Gaussian video reconstruction. Zhenyang Li[†], **Xiaoyang Bai[†]**, Tongchen Zhang, Pengfei Shen, Edmund Y. Lam, Yifan Peng. In *proceedings of IEEE International Symposium on Mixed and Augmented Reality (ISMAR)* 304-314 (2025)

Learning RGBD imaging via asymmetrically focused stereo cameras. Liangxun Ou, Yuhui Liu, **Xiaoyang Bai**, Yifan Peng. In *The Visual Computer* 41.15 (2025)

Learning a global underwater geolocalization model with Sectoral Transformer. **Xiaoyang Bai**, Zhongmin Zhu, Alexander Schwing, David Forsyth, Viktor Gruev. In *Optics Express* 32.12 (2024)

Polarization-based underwater geolocalization with deep learning. **Xiaoyang Bai**, Zuodong Liang, Zhongmin Zhu, Alexander Schwing, David Forsyth, Viktor Gruev. In *eLight* 3.1 (2023)

Angle of polarization calibration for omnidirectional polarization cameras. **Xiaoyang Bai**, Zhongmin Zhu, Alexander Schwing, David Forsyth, Viktor Gruev. In *Optics Express* 31.4 (2023)

Services

Conference Organizer

- Program Manager of Mixed Reality Program, ACM SIGGRAPH Asia 2025
- Technical Paper Committee of ICXR 2025
- Program Committee of Emerging Technology Program, ACM SIGGRAPH Asia 2026

Journal Reviewer

- ACM Transactions on Graphics (TOG)
- Computer Visual Media Journal (CVMJ)
- IEEE Transactions on Visualization and Computer Graphics (TVCG)
- Computer and Graphics

Conference Reviewer

- ACM SIGGRAPH, 2025&2026
- IEEE Conference on Virtual Reality (VR), 2025&2026
- IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2026
- IEEE International Conference on Computational Imaging (ICCP), 2026